

Technology Stack

Date	06 July 2026
Project Name	Rising Waters
Maximum Marks	2 Marks

Technology Stack :

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Creates a responsive and user-friendly web interface for entering weather data and viewing flood prediction results.
2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, and connects the machine learning model with the frontend.
3	Database / Data Storage	CSV Dataset, Joblib (.save files)	Historical flood dataset is stored in CSV format, while the trained model and scaler are stored using Joblib for prediction.
4	Cloud / Hosting / Deployment	Localhost (Flask Development Server)	Used to develop and test the application locally. Can be deployed to cloud platforms in the future.
5	Version Control & CI/CD	Git, GitHub	Used to manage source code, track changes, and share the project repository.
6	Third-Party APIs / Other Tools	XGBoost, Scikit-learn, Pandas, NumPy	XGBoost performs flood prediction, Scikit-learn is used for preprocessing, and Pandas/NumPy handle data processing and analysis.